Rule Book

AstralWeb Innovate

Problem Statement

Develop an interactive, highly responsive web platform titled "Project Zenith: The Celestial Eye" that functions as a real-time cosmic radar. The platform must allow users to select any geographic coordinate on Earth and dynamically display the exact celestial bodies ranging from the International Space Station and active satellites to distant planets and constellations currently passing through that location's zenith. To achieve this, participants must seamlessly integrate real-time telemetry APIs, such as NASA Horizons, OpenNotify, and CelesTrak, with advanced front-end mapping libraries like CesiumJS, Leaflet, React, and Next.js. The core challenge lies in moving beyond static web design to focus on precise geospatial mapping, live data synchronization, and astronomical accuracy. Ultimately, your task is to blend scientific rigor with creative UX/UI design to deliver an intuitive, cross-device compatible, and immersive educational experience featuring real-time animations and interactive components.

Event Overview

- Event Venue: Online
- Dates: 10th June 2026 – 30th June 2026
- Reporting Time:
- Inaugural Ceremony – 10th June 2026
- Valedictory Ceremony – 30th June 2026
- Participation Type: Individual as well as Team

Competition Format

Round 2: Web Submission

In this primary coding phase, teams must transition their theoretical blueprints into a fully functional, hosted web application. The objective is to build an interactive platform where users can select any geographic coordinate on Earth and instantly visualize the celestial bodies such as the ISS, active satellites, planets, and constellations that are currently overhead in real time. Participants are required to submit a live hosted URL alongside a public GitHub repository link containing a comprehensive README file that explains the project's setup, dependencies, and functionality. Key deliverables for this sprint include a highly responsive UI utilizing advanced CSS (such as Grid or Flexbox) for cross-device compatibility, a functional 3D globe or interactive map for coordinate capture, and seamless, dynamic real-time data fetching. Submissions will be rigorously evaluated based on their UI/UX aesthetics and thematic consistency, the accuracy and smoothness of the real-time data integration, overall code quality, and feature richness, which includes creative additions like orbit paths, speed trackers, or constellation overlays. Top-performing teams will subsequently advance to the final live audit round.

- Starting Date: After Round 1 results are announced.
- Deadline: 26th June 2026, 11:59 PM
- Submission Format: Submit a live hosted URL and a public GitHub repository link. Include a README file explaining setup and functionality.

Key Deliverables:

- Interactivity: A functional interactive map or 3D globe that captures user-selected coordinates.

- Real-time Data Fetching: Dynamic display of celestial bodies currently above the selected geographic location.
- Responsive UI: Application must use advanced CSS (Grid/Flexbox) to ensure a high-quality experience on mobile, tablet, and desktop.
- Code Quality: Clean, well-documented code with a clear README.

Evaluation Criteria:

- UI/UX Aesthetic & Cosmic Theme: Visual appeal, navigation, and thematic consistency.
- Real-time Data Integration: Accuracy and smoothness of live celestial data display.
- Feature Richness: Creative extras such as orbit paths, speed trackers, or constellation overlays.
- Code Structure & Documentation: Cleanliness and quality of codebase and README.

Outcome:

- Top-performing teams will advance to Round 3. Teams will receive feedback on design, data integration, and technical implementation.

Submission Requirements

Round 2: Website Submission

- Live Hosted URL: A working link to the deployed web application.
- Public GitHub Repository: Full codebase, assets, and project files.
- README File: Explains functionality, setup instructions, API keys required, and dependencies.
- Design Requirements: Must implement the blueprint concept, be fully responsive, and include real-time data features.